Time-horizon of decision making

Short vs. long run

one of the inputs is fixed.

This denotes the time horizon over which a firm is making decisions.

machines

capital (K) constant (fixed)

unable to change.

Labor = not fixed (variable)

L, K fixed

Time-horizon of decision making

Short vs. long run

This denotes the time horizon over which a firm is making decisions.

For example - if we are to study short run responses to changes in external conditions, then certain inputs (capital) might have to be held constant. If the time interval under consideration is one week, then the size of Tesla's battery manufacturing plant has to be considered constant. However, in the long run, the size of the plant can also change.

In line with this example, in this course we consider the difference between short and long run to be that capital is held constant.

Does this mean firms have no flexibility in short run?

Does this mean firms have no flexibility in short run?

Constrained

No. Although firms cant change the amount of capital, they can change other inputs, which in our case is labor (L). Continuing in the same example, Tesla can hire/fire workers from their factories at relatively short notice (depending on contracts), although the number of car-assembly unit is unlikely to change in the short run.

Note that there might be other inputs in the production process that may (or may not) vary in the short run. In this course, we consider the simple two-input model with labor and capital, where capital does not change in the short run.

Types of short-run costs

Types of short-run costs

Fixed costs

These are the costs associated with the fixed factors of production, namely capital (K). These costs are independent of the amount of production – that is, irrespective of how much the firm produces, the fixed costs are ***constant*** in the short run.

Types of short-run costs

long-run

Fixed costs
These are the costs associated with the fixed factors of production, namely capital (K). These costs are independent of the amount of production – that is, irrespective of how much the firm produces, the fixed costs are ***constant*** in the short run.

$\boxed{\textbf{Variable costs}}$
These are short-run costs that are associated with the inputs that can be changed in the short run, which in our case is labor (L). In our previous example, although Tesla cannot add new machines to its factory, it could potentially add to its output by hiring more workers – the cost entailed to hire these workers would be a part of the variable costs of production.

Average costs

New (but related) concepts – short run costs (variable and fixed costs)

$$AC = \frac{TC}{Q}$$

Average costs

New (but related) concepts – short run costs (variable and fixed costs)

$$\text{Average variable cost (AVC)} = \frac{\text{Total variable costs}}{\text{Quantity}} = \frac{TVC}{q}$$

Average costs

New (but related) concepts – short run costs (variable and fixed costs)

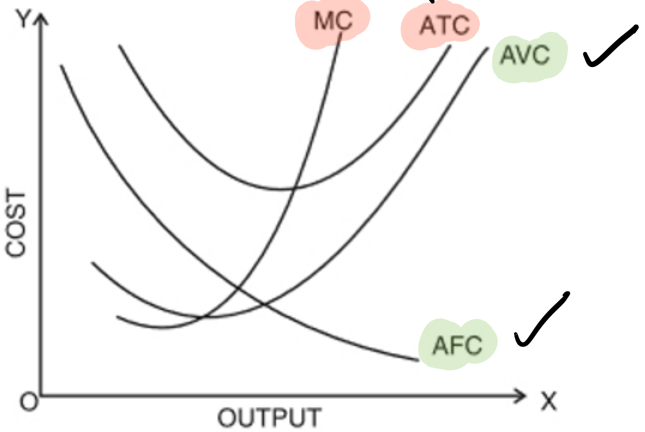
$$\text{Average variable cost (AVC)} = \frac{\text{Total variable costs}}{\text{Quantity}} = \frac{TVC}{q}$$

$$\text{Average fixed cost (AFC)} = \frac{\text{Total fixed costs}}{\text{Quantity}} = \frac{TFC}{q}$$

Average fixed and variable costs

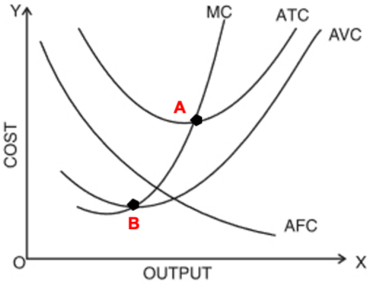
Graphical representation

link



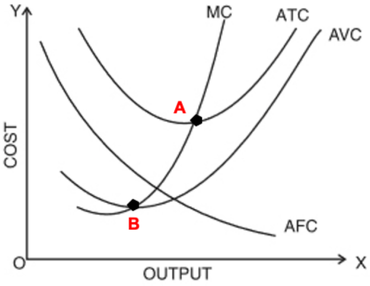
Average fixed and variable costs

Some stylized facts (or properties)



Average fixed and variable costs

Some stylized facts (or properties)



- ATC = AFC + AVC (always true!)

$TC = TFC + TVC$

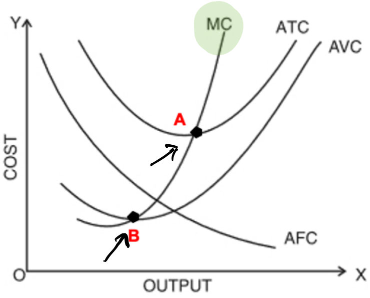
Divide by Q

$\frac{TC}{Q} = \frac{TFC}{Q} + \frac{TVC}{Q}$

$ATC = AFC + AVC$

Average fixed and variable costs

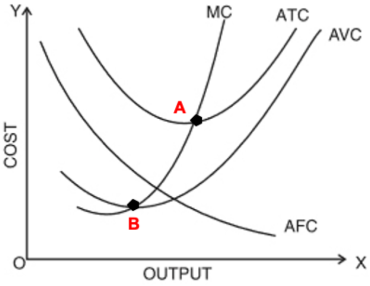
Some stylized facts (or properties)



- ATC = AFC + AVC (always true!)
- MC intersects ATC and AVC at their minimum values (points A and B)

Average fixed and variable costs

Some stylized facts (or properties)

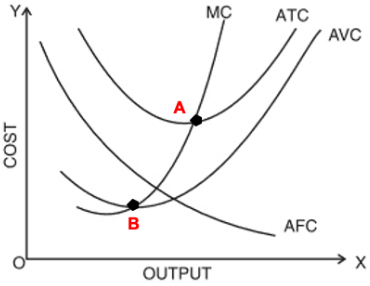


- ATC = AFC + AVC (always true!)
- MC intersects ATC and AVC at their minimum values (points A and B)
- When MC < AVC, then AVC is falling (left of point B).

When MC < AC, then AC is falling.

Average fixed and variable costs

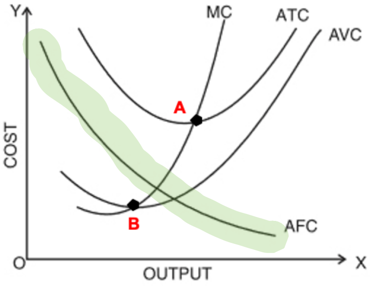
Some stylized facts (or properties)



- ATC = AFC + AVC (always true!)
- MC intersects ATC and AVC at their minimum values (points A and B)
- When MC $<$ AVC, then AVC is falling (left of point B).
- When MC $>$ AVC, then AVC is rising (right of point B).

Average fixed and variable costs

Some stylized facts (or properties)



$AFC = \frac{TFC}{Q \uparrow}$

- ATC $=$ AFC $+$ AVC (always true!)
- MC intersects ATC and AVC at their minimum values (points A and B)
- When MC $<$ AVC, then AVC is falling (left of point B).
- When MC $>$ AVC, then AVC is rising (right of point B).
- AFC is always falling. AFC $= \frac{TFC}{q}$. Since TFC is constant, when q rises then AFC $\downarrow$.

Question: Will a firm always be able to use the cost minimizing combination of L and K in the short run?

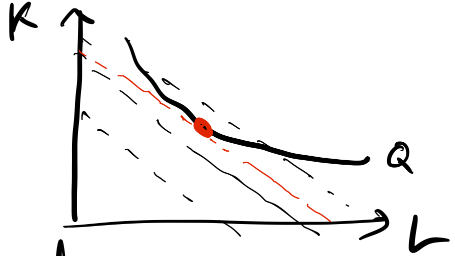
Try yourself, and then check out the solution!

cost min.

↓

tangency cond.

b/w isocost and isoquant



Question: Will a firm always be able to use the cost minimizing combination of L and K in the short run?

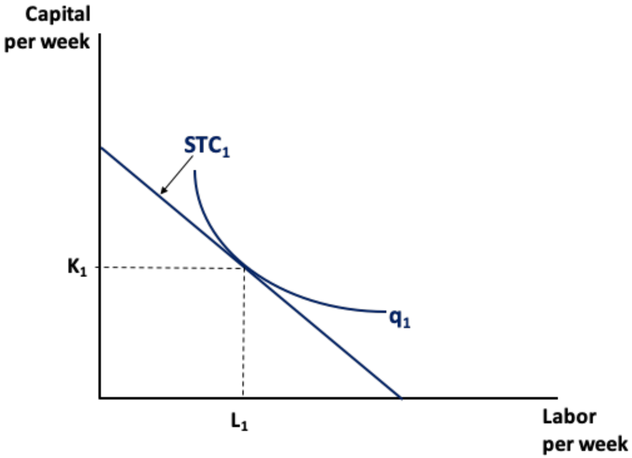
Try yourself, and then check out the solution!

The answer is **No**.

The reason is that the firm does not have the flexibility to choose any level of capital (fixed factor of production) in the short run. It is constrained to stick with the same amount of capital.

Firm decision in the short run

Suppose the firm is currently producing q_1 units of output using L_1 units of labor and K_1 units of capital.

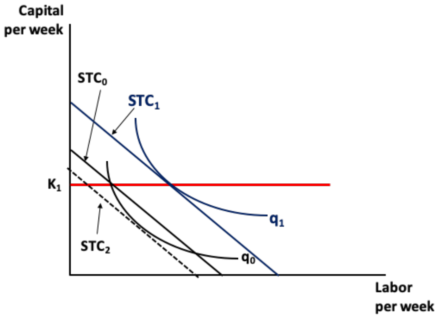


Firm decision in the short run

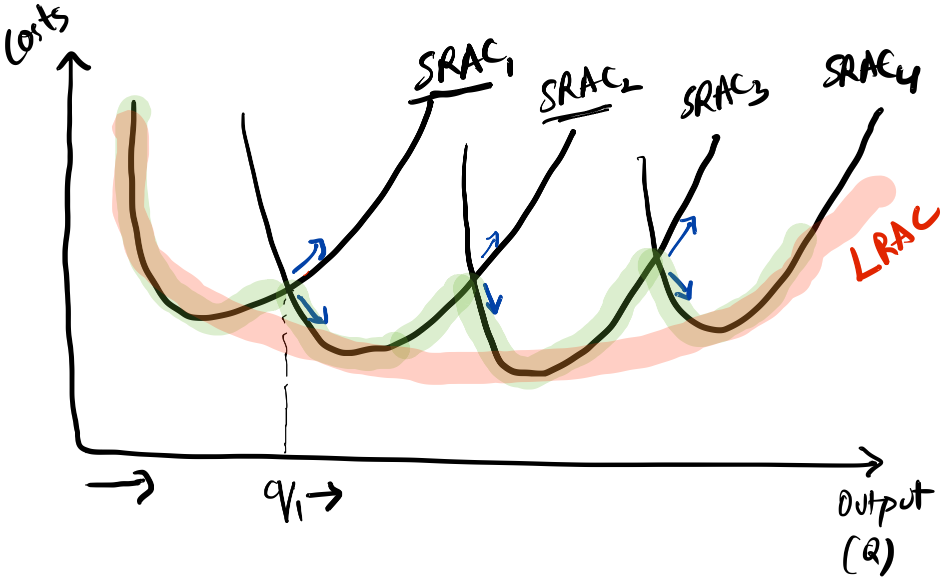
Not always able to choose the cost-minimizing level of inputs.

Now, if the firm wants to produce q_0 units of output, where $q_0 < q_1$. Then the firm will still be using K_1 units of capital (as it is fixed), and will change the units of labor (variable factor) to L_0.

Note that (L_0, K_1) is not a cost-minimizing input combination for producing output q_0 (tangency condition is not satisfied).

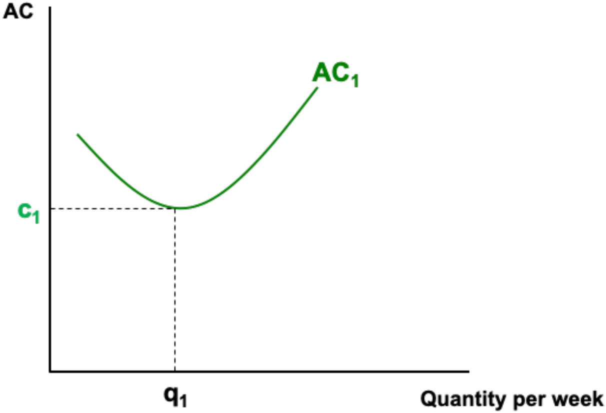


Deriving long run AC from short run AC

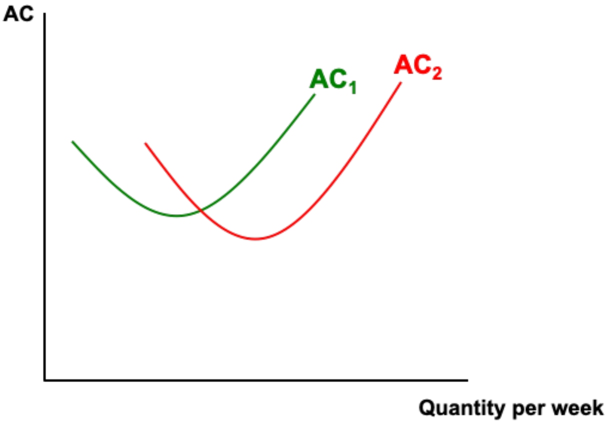


Deriving long run AC from short run AC

In the short run, the level of capital is fixed, so we operate on a given average cost curve. Therefore, the minimum cost of producing quantity q_1 is c_1.

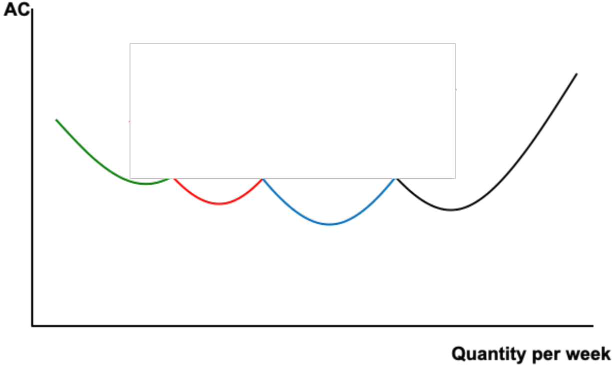


Deriving long run AC from short run AC



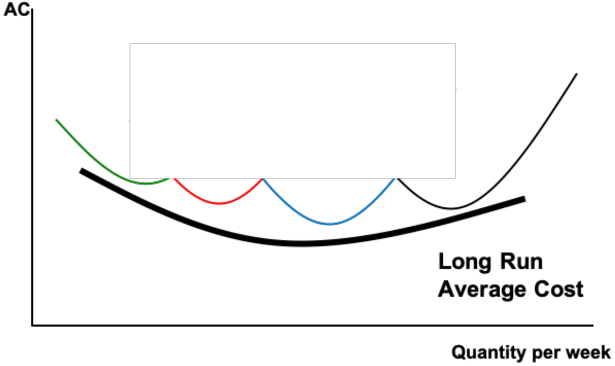
Deriving long run AC from short run AC

Therefore, the long run AC curve is given by the lower envelope of different short run AC curves.



Deriving long run AC from short run AC

Therefore, the long run AC curve is given by the lower envelope of different short run AC curves. In this course, we use a "smoothened" version of this curve.



Deriving long-run total costs $\rightarrow$ TC, AC, MC.

Production function: $\boxed{q = K.L}$

w, v = constants

q = number.

$TC = TC(q)$

$TC = wL + vK$

optimum L, K for a given q

$MP_L = \frac{\partial}{\partial L}[q] = \frac{\partial (K \cdot L)}{\partial L} = K$

$MP_K = \frac{\partial}{\partial K} q = \frac{\partial (K \cdot L)}{\partial K} = L$

$\frac{MP_L}{MP_K} = \frac{w}{v}$

$\Rightarrow \frac{K}{L} = \frac{w}{v}$

$\Rightarrow L = \frac{Kv}{w}$

$q = K \cdot L$

$\Rightarrow q = K \cdot \left(\frac{Kv}{w}\right) \Rightarrow K^2 = \frac{qw}{v} \Rightarrow K = \left(\frac{qw}{v}\right)^{1/2}$

Deriving long-run total costs

$K = \left(\frac{qw}{v}\right)^{1/2}$ and $L = \frac{Kv}{w}$

Production function: $q = K.L$

$$L = \left(\frac{qw}{v}\right)^{1/2} \cdot \frac{v}{w} = \left(\frac{qv}{w}\right)^{1/2}$$

$$TC = wL + vK$$

$$= w\left(\frac{qv}{w}\right)^{1/2} + v\left(\frac{qw}{v}\right)^{1/2}$$

$$= (wqv)^{1/2} + (wqv)^{1/2}$$

$$= 2(wqv)^{1/2}$$

$$AC = \frac{TC}{q} = 2(qvr$$

Example: Short-run costs

Try it yourself!

A firm producing hockey sticks has the following production function: $q = 2\sqrt{KL}$. In the short run, the firm's capital is fixed at K = 100. The rental rate of K is $v = \$1$ and the wage rate for L is $w = \$4$.

a. Calculate the firm's short-run total cost and average cost functions.

b. The firm's short run marginal cost function is given by SMC = q/50. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

c. Graph the SAC and the SMC curves for the firm. Indicate the points in part (b).

d. Where does the SMC curve intersect the SAC curve? Explain why the SMC curve will always intersect the SAC at its lowest point.

Example

a. Calculate the firm's short-run total cost and average cost functions.

Part a.

We know that $q = 2\sqrt{KL}$ and that K=100. Therefore,
$q = 2\sqrt{100L} \implies q = 20\sqrt{L} \implies \sqrt{L} = \frac{q}{20} \implies L = \frac{q^2}{400}$.

Therefore, $STC = vK + wL = 100 + \frac{q^2}{100}$, and

$SAC = \frac{STC}{q} = \frac{100}{q} + \frac{q}{100}$

Example

b. The firm's short run marginal cost function is given by SMC $= q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Example

b. The firm's short run marginal cost function is given by $SMC = q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Part b.

We know that $SMC = q/50$

If $q = 25$, then $STC = 106.25$,

Example

b. The firm's short run marginal cost function is given by SMC $= q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Part b.

We know that SMC $= q/50$

If $q = 25$, then STC $= 106.25$, SAC $= 4.25$

Example

b. The firm's short run marginal cost function is given by SMC $= q/50$. What are the STC, SAC and SMC for the firm if it produces 25, 50, 100 and 200 hockey sticks?

Part b.

We know that SMC $= q/50$

If $q = 25$, then STC $= 106.25$, SAC $= 4.25$ and SMC $= 0.5$

If $q = 50$, then STC $= 125$, SAC $= 2.5$ and SMC $= 1$

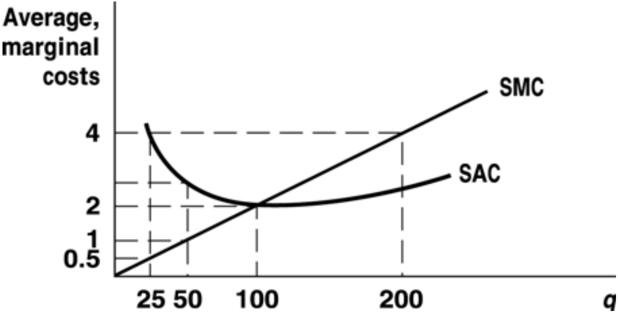
If $q = 100$, then STC $= 200$, SAC $= 2$ and SMC $= 2$

If $q = 200$, then STC $= 500$, SAC $= 2.5$ and SMC $= 4$

Example

c. Graph the SAC and the SMC curves for the firm. Indicate the points in part (b).

Part c.



Example

d. Where does the SMC curve intersect the SAC curve? Explain why the SMC curve will always intersect the SAC at its lowest point.

Part d.

SMC and SAC intersect at $q = 100$. As long as the marginal cost of producing one more unit is below the average cost curve, average costs will be falling. Similarly, if the marginal cost of producing one more unit is higher than the average cost, then average costs will be rising. Therefore, the SMC curve must intersect the SAC curve at its lowest point.